Supplementary Information:

Context Without Contact: Top-Down Information Flow from Shared Modulation in Uncoupled Prebiotic Systems

This PDF file includes:

Supporting Figures S1 to S3

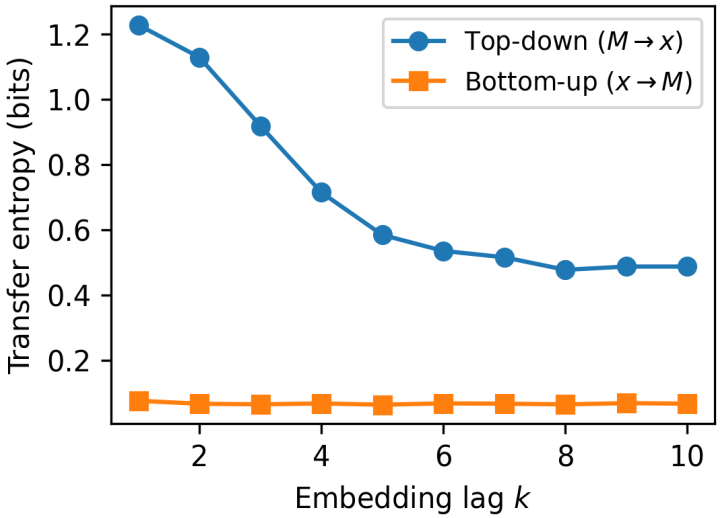

Figure S1. Mean transfer entropy (TE) as a function of embedding lag k for modulation amplitude A=25. Top-down TE (M→x, circles) measures information from the macroscopic mean field to an individual unit, while bottom-up TE (x→M, squares) measures the reverse direction. Values are averaged over 10 random seeds. Parameters: N=1000, K■=100, $r_i \in [3.9,4.0]$, T=10,000 steps (first 1,000 discarded), 15-bin histograms.

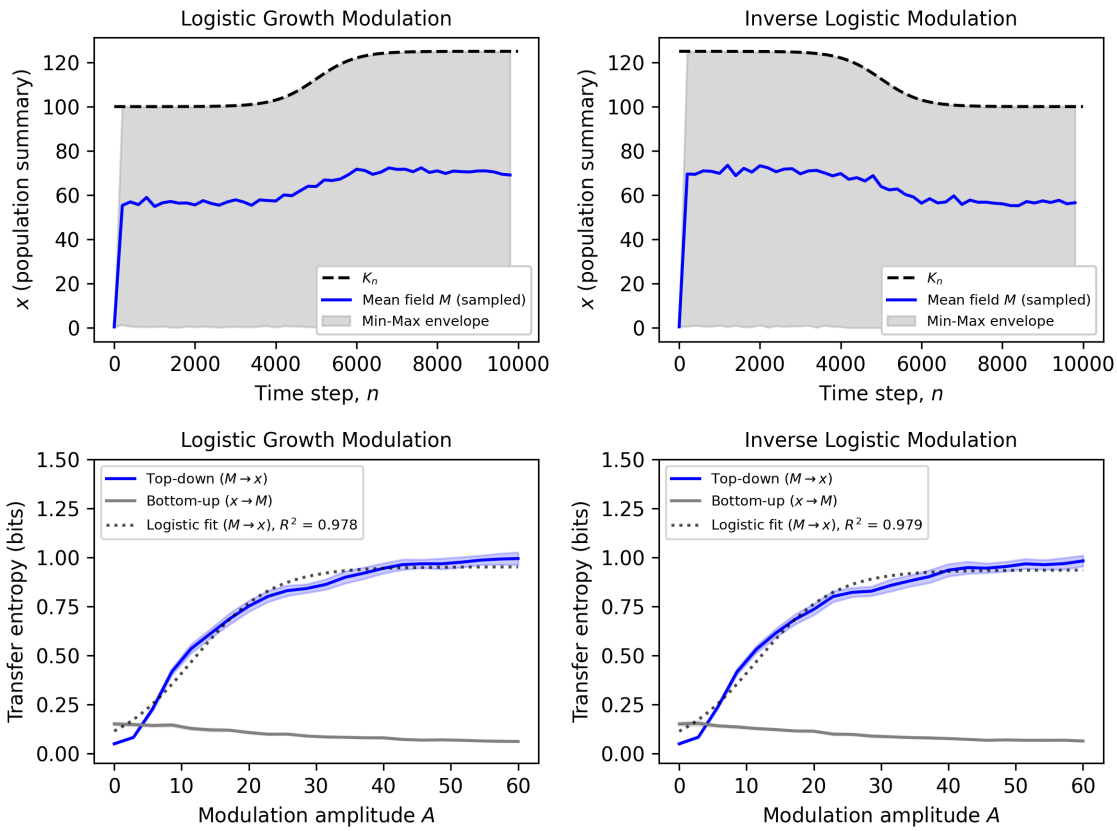

Figure S2. Directional information flow from macroscopic to microscopic dynamics under different modulation profiles. Transfer entropy (TE) between the mean field M_n and a representative unit x_n, computed in both directions: M → x (blue) and x → M (gray), for a population of N = 1000 uncoupled logistic units evolving under carrying capacity modulated by (A,C) logistic increase, and (B,D) inverse logistic. In all cases K_0 = 100, A = 25, $r_i \in$ [3.9, 4.0], and steepness is k = 0.05, simulated over 10 independent simulations, each run for T = 10,000 steps (first 1,000 discarded). Top row: The black curve shows the external modulation K_n, the blue line corresponds to the mean field M, sampled every 200 steps, and the shaded gray band indicates the range of individual states at each sampled point. Bottom row: TE estimated using a histogram-based method with lag k = 2 and 15 bins. Shaded regions indicate 95% confidence intervals.

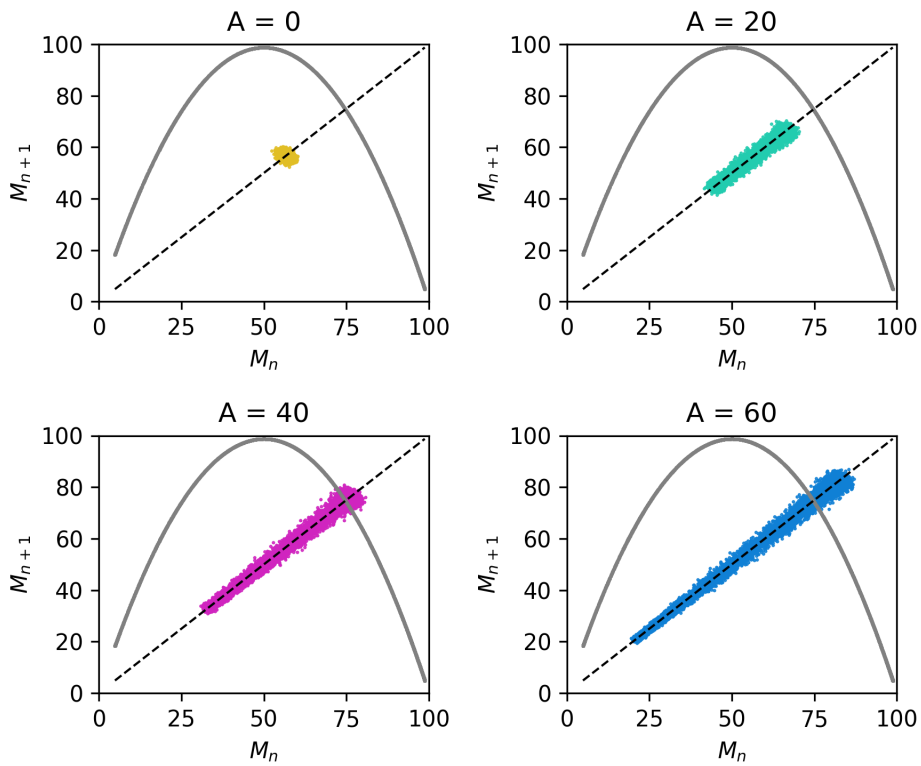

Figure S3. Structural modulation induces low-dimensional macroscopic dynamics. Return maps of (M_n, M_{n+1}) for A = {0, 20, 40, 60}, N = 1,000, K_0 = 100, $r_i \in$ [3.9, 4.0], and T = 10,000 steps (first 1,000 discarded). The dashed line shows the identity $M_{n+1} = M_n$ and the gray curve shows the single-unit logistic map for comparison. At A=0, points cluster near ($K_0/2$, $K_0/2$). As A increases, shared modulation widens the distribution of M_n, producing a diagonal spread.